

Fraudulent Job Posting Detection System

An End-to-End ML + NLP Risk Screening Platform

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What This System Does

A machine-learning system that analyzes job advertisements, estimates fraud probability, identifies risk signals, validates input quality, and generates an explainable risk assessment — from raw text to a formal PDF report.

Who This Is For

Built as a portfolio demonstration for data science recruiters, ML interviewers, and hiring managers evaluating complete ML lifecycle understanding — from data through productization.

Core Capability

Combines NLP text analysis, structured job metadata, gradient boosting, rule-based risk signals, and input validation into a single decision-support application.

99.02%

Accuracy

97.92%

Precision on Fraud

88.96%

F1 Score

99.53%

ROC-AUC

Fake Job Posts Leave Signals — But They Are Not Always Obvious

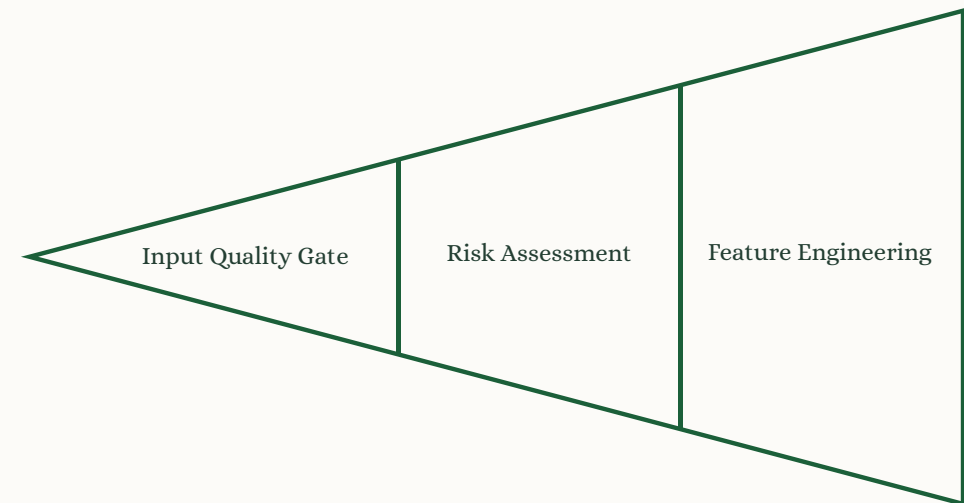
The Problem

Fraudulent job postings deliberately mimic legitimate advertisements. A simple keyword filter is insufficient — legitimate and fraudulent postings often share similar characteristics, making automated detection genuinely hard.

- Exploit job seekers with false opportunities
- Request payments, fees, or deposits
- Collect sensitive personal information
- Redirect applicants to suspicious channels
- Advertise unrealistic employment terms

The Solution Architecture

Combine NLP + Structured Metadata + Machine Learning + Rule-Based Risk Signals + Input Validation into a layered decision-support system.



This is a **decision-support system**, not definitive proof of fraud. The pipeline is designed to flag, explain, and report — giving analysts the information they need to act.

17,880 Job Postings — Only 4.84% Are Fraudulent

Dataset Overview

17,880

Total Records

18 features per posting

866

Fraudulent

4.84% of all postings

17,014

Legitimate

95.16% of all postings

⚠ Severe class imbalance (19.64:1 ratio) makes standard accuracy a misleading metric. F1 score and ROC-AUC are the primary evaluation criteria.

Key EDA Findings

Company Profile Presence

Legitimate postings had **84.01%** with company profiles vs. only **32.22%** for fraudulent postings — a strong structural signal.

Salary Information

15,012 postings are missing salary data; only **2,868** include salary ranges. Salary absence alone is not a reliable fraud signal given base rates.

Additional Dimensions Explored

Industries, job functions, employment types, job titles, requirements depth, suspicious word frequency, and company information completeness.

These are dataset-level patterns, not universal fraud rules. EDA informed feature selection, not hard-coded thresholds.

The Model Combines Language With Job Metadata

Feature Engineering Pipeline — 7,627 Total Features

Text Features

5,000 TF-IDF dimensions

- Fields: title, company profile, description, requirements, benefits
- HTML/entity cleaning, URL/email/phone placeholders
- Lowercasing, punctuation removal
- Config: max_features=5,000, ngram_range=(1,2), min_df=2, max_df=0.95, English stopwords

Categorical Features

2,616 encoded dimensions

- OneHotEncoder applied to:
- employment_type, required_experience, required_education
- industry, function, country, state, city

Numerical Features

5 dimensions

- Description length
- Requirements length
- Company profile length
- Benefits length
- Suspicious word count

Binary Features

6 dimensions

- Telecommuting flag
- Company logo present
- Screening questions present
- Salary availability
- Company profile present
- Requirements present

Model Development & Hyperparameter Tuning

Models Evaluated

Seven algorithms were benchmarked using stratified cross-validation before selecting the final model:

01

Logistic Regression

Linear baseline; fast and interpretable

02

Linear SVM

Strong for high-dimensional TF-IDF spaces

03

Multinomial Naive Bayes

Probabilistic text classification baseline

04

Random Forest

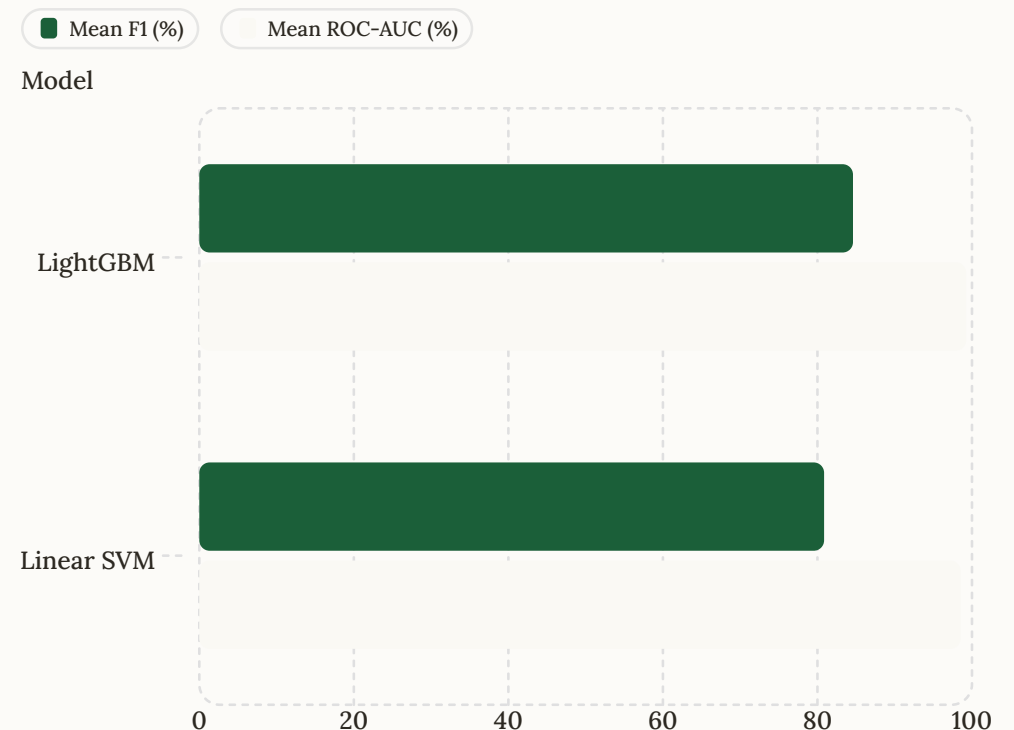
Ensemble of decision trees; robust to noise

05

XGBoost / LightGBM / CatBoost

Gradient boosting variants; best for mixed feature spaces

Cross-Validation Results (Top 2)

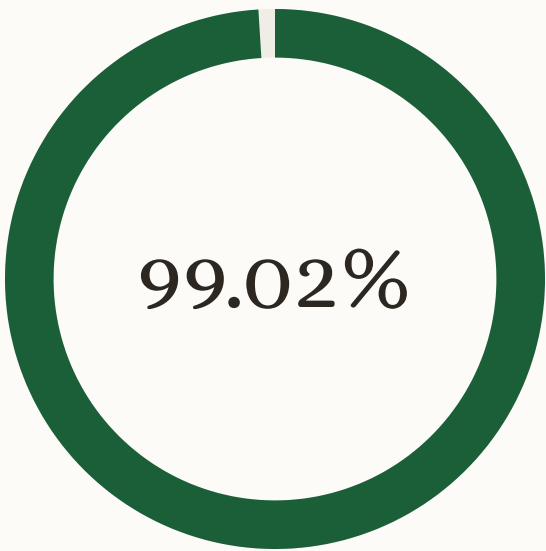


Optimization Strategy

- **Split:** Stratified 80/20 train-test (random_state=42) to preserve class proportions
- **Tuning:** RandomizedSearchCV – 10 iterations, 3-fold CV, F1 scoring
- **Imbalance handling:** LightGBM scale_pos_weight = 19.64 (legitimate-to-fraud ratio)
- **Best CV F1 after tuning:** 82.59%

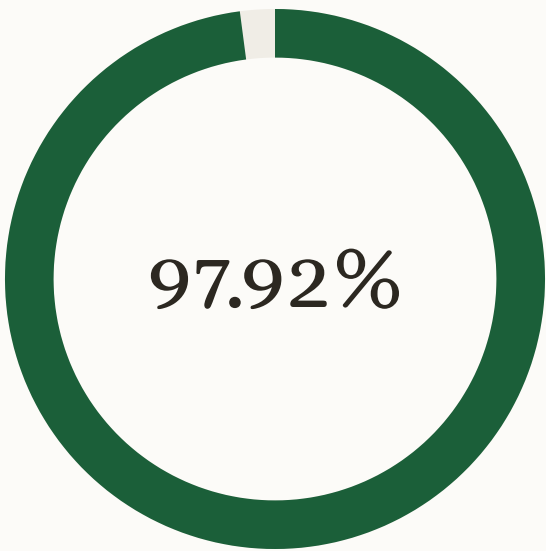
❏ LightGBM selected for superior fraud-class balance, ROC-AUC, and native support for class imbalance weighting.

99.02% Accuracy — With 97.92% Precision on Fraud



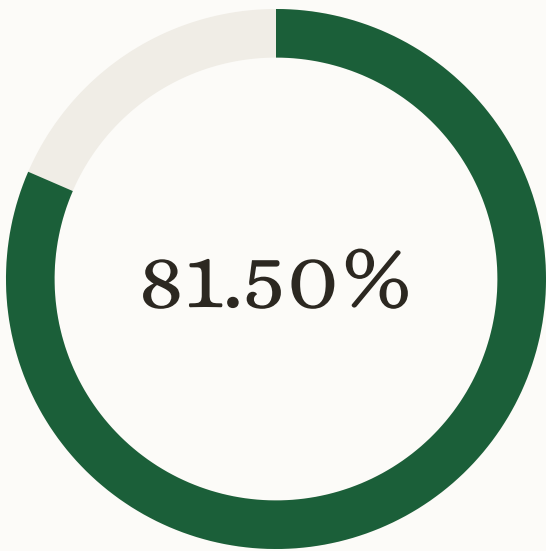
Accuracy

Overall correct classifications on held-out test set



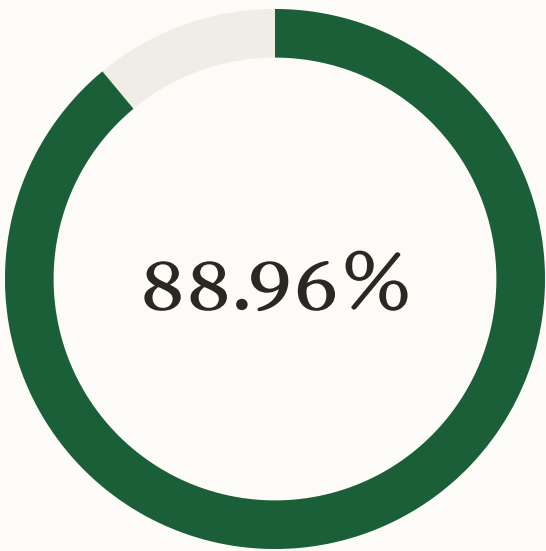
Precision

Of flagged fraud, 97.92% were truly fraudulent



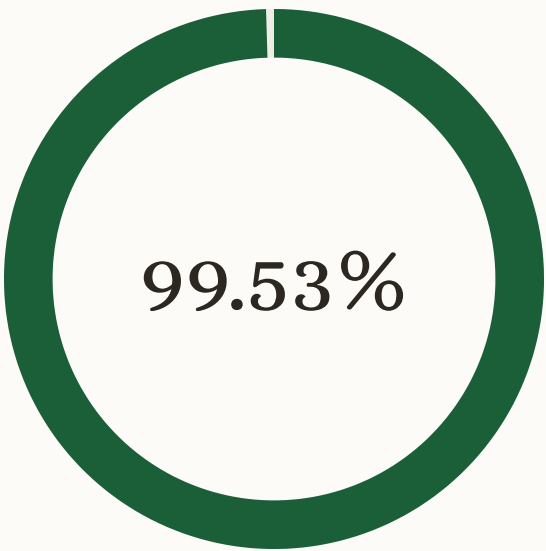
Recall

Of all actual fraud, 81.50% were caught



F1 Score

Harmonic mean of precision and recall for fraud class



ROC-AUC

Near-perfect discrimination between classes

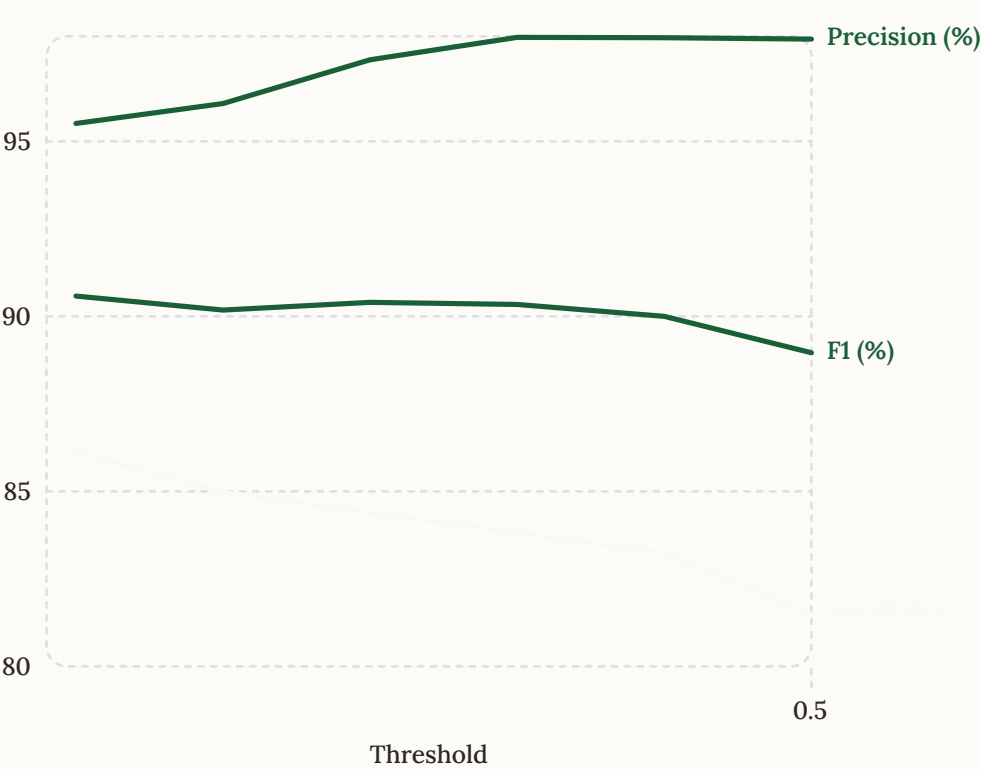
Class-Level Breakdown

Class	Precision	Recall	F1 Score
Legitimate	99.07%	99.91%	99.49%
Fraudulent	97.92%	81.50%	88.96%

Fraud detection performance matters more than accuracy alone because the dataset is highly imbalanced. A model predicting all-legitimate would achieve 95.16% accuracy — the fraud-class metrics are the meaningful signal.

The Decision Threshold Changes the Cost of Missing Fraud

Threshold Optimization Analysis



Threshold Selection Rationale

Application Threshold: 0.35

Selected because it maximizes F1 (90.40%) while maintaining very high precision (97.33%), controlling false alarms without sacrificing too many true fraud detections.

Trade-off Insight

Lowering threshold to 0.25 improves recall to 86.13% but reduces precision to 95.51%. In fraud detection, false negatives (missed fraud) and false positives (wrongly flagging legitimate postings) carry different costs.

Error Analysis at 0.35

32

False Negatives

Fraudulent postings missed by the model

3

False Positives

Legitimate postings incorrectly flagged

Fraud detection is a trade-off, not a perfect binary decision. Error analysis demonstrates production-level thinking about model behavior beyond aggregate metrics.

The System Doesn't Just Predict — It Checks and Explains

Rule-Based Risk Signals

Alongside ML probability, rule-based indicators escalate or contextualize the risk level:

- **Payment-related:** registration fee, processing fee, security deposit, payment required, bank account requests, wire transfers, cryptocurrency, gift cards
- **Communication red flags:** WhatsApp-only, Telegram-only contact
- **Unrealistic terms:** guaranteed income, unrealistic earnings, artificial urgency, no experience required

Risk Level Output: ML Probability + Risk Signals →

MINIMAL	LOW
ELEVATED	HIGH
CRITICAL	

Risk signals support assessment; they are not definitive proof of fraud.

Input Quality Gate

Before model inference, the application validates content meaningfulness. This production-minded feature prevents inference on meaningless or adversarial input.

Validation Checks

- Empty or very short fields
- Insufficient vocabulary or word count
- Gibberish and abnormal word lengths
- Repeated characters or sentences
- Excessive punctuation or numbers
- Abnormal entropy, sparse content

Validation Techniques

- Shannon entropy analysis
- Vocabulary and pronounceability checks
- Vowel/consonant ratio analysis
- Unusual letter pattern detection

1

Invalid Input

→ **BLOCK** with explanation

2

Valid Input

→ **MODEL INFERENCE**

The ML Model Became a Full Interactive Screening Application

Streamlit Application Workflows



Single Analysis

Analyze an individual posting with full risk assessment



Batch Analysis

Process multiple postings from a CSV upload



Compare

Side-by-side comparison of two postings



Real or Fake?

Interactive prediction game for exploration



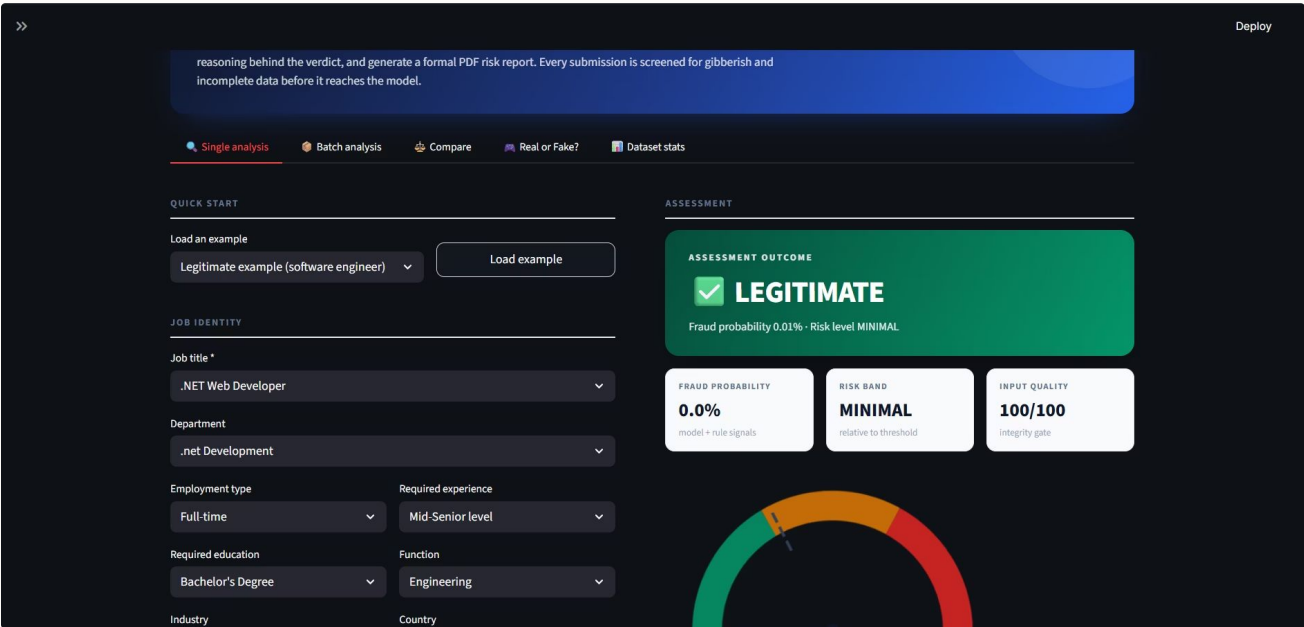
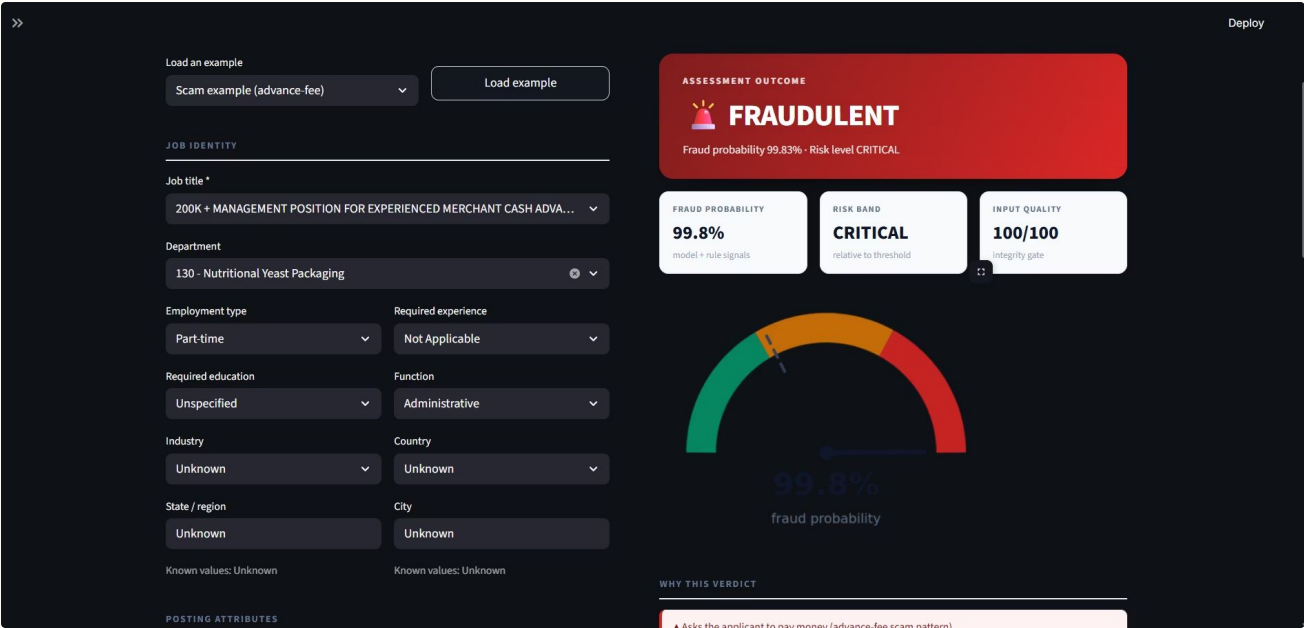
Dataset Stats

Explore dataset-level patterns and distributions

Single Analysis Outputs

- Verdict: FRAUDULENT / LEGITIMATE
- Fraud probability score
- Risk level: MINIMAL → CRITICAL
- Input quality assessment
- Risk indicators triggered
- Explanation and recommendations

Application Screenshots



From Prediction to an Actionable Risk Report

Automated PDF Reporting

The application generates formal PDF risk reports using **ReportLab**. Each report is a complete, standalone audit artifact containing:

- 1

Report Header

Unique report identifier, timestamp, job title, and location
- 2

Prediction Summary

Fraud prediction, probability score, risk level, and input quality assessment
- 3

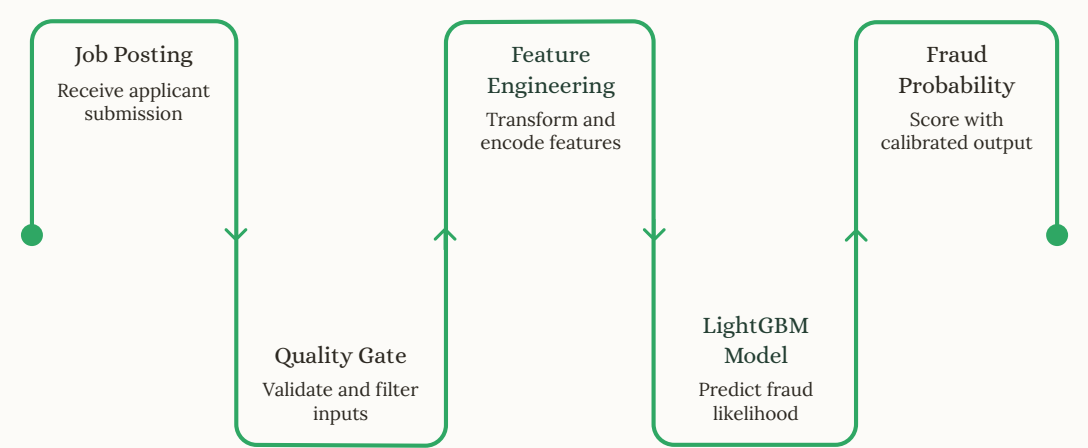
Risk Narrative

Structured risk assessment explanation with triggered indicators
- 4

Full Job Details

Complete description, requirements, benefits, and company profile text

End-to-End Pipeline



PDF Report Preview

FRAUDULENT JOB POSTING DETECTION SYSTEM

Report FJD-ADB11AFC

Job Posting Risk Assessment Report

Machine-learning assessment of the likelihood that the submitted job advertisement is fraudulent.

Report identifier	FJD-ADB11AFC
Generated (UTC)	2026-08-25 14:10:39 UTC
Generated (local)	2026-08-25 19:40:39 (local)
Job title	200K + MANAGEMENT POSITION FOR EXPERIENCED MERCHANT CASH ADVANCE REPS
Location	Not specified

ASSESSMENT OUTCOME

FRAUDULENT

Fraud probability 99.83% · Risk level CRITICAL · Input quality 100/100

1. Executive summary

The submitted advertisement for the position "200K + MANAGEMENT POSITION FOR EXPERIENCED MERCHANT CASH ADVANCE REPS" was assessed as **FRAUDULENT** with a fraud probability of 99.83%, placing it in the **CRITICAL** risk band. The posting exhibits patterns consistent with fraudulent recruitment activity and should not be acted upon without independent verification of the employer.

99.8%
fraud probability

Figure 1 — Fraud probability against the configured decision threshold.

2. Risk assessment

From Job-Posting Classification Toward a Recruitment Trust & Safety Platform

What This Project Demonstrates

Data Science & NLP

- EDA, class imbalance analysis, dataset-level pattern discovery
- TF-IDF feature extraction, text preprocessing, HTML/entity cleaning

Feature Engineering

- Text, categorical, numerical, and binary feature integration
- 7,627 total features across four feature families

Machine Learning & Evaluation

- Multi-model evaluation, LightGBM selection, stratified cross-validation
- Precision, recall, F1, ROC-AUC, threshold optimization, error analysis

ML Engineering & Productization

- Rule-based risk signals, interpretable decision support, input validation
- Streamlit application, multiple workflows, ReportLab PDF generation

Future Roadmap



Model Explainability

SHAP explanations for per-prediction model transparency



Transformer NLP

BERT-based semantic understanding for richer text signals



Multilingual Support

Global job market coverage beyond English postings



Employer Verification

Reputation scoring, URL/domain reputation integration



MLOps & Scaling

Model monitoring, drift detection, automated retraining, cloud deployment

Contact

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